

AYŞE CEREN ŞENEL

Computer Engineer

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EDUCATION

Ankara University

Computer Engineering

- Projects: aysecerensenal.com 🔗
- GPA: 3.30 / 4.00

09/2023 – Present

Meram Anatolian High School

Grade: 96.69 / 100

09/2018 – 06/2022

SUMMARY

Computer Engineering student at Ankara University with research and software development experience in Artificial Intelligence (RAG, NLP), Post-Quantum Cryptography (PQC), Multimodal Signal Processing (EEG-fMRI), and Embedded/System Architectures.

Currently working on decision support systems, quantum algorithms, OSINT platforms, and national cybersecurity transformation strategies within advanced research projects conducted at Boğaziçi University MIMLab, Microsoft (Azure Quantum), and the Directorate of Communications of the Presidency of the Republic of Türkiye. Holding leadership roles in Teknofest teams and serving as a communication researcher in the OTUSAT CubeSat project while engineering modular software architectures and data processing pipelines.

My goal is to play an active role in architecting innovative, high-impact systems centered around AI, quantum software development, and defense/cybersecurity technologies.

TECHNICAL SKILLS

Programming Languages

Python, TypeScript, JavaScript, SQL, C, C++, Java, Q# (Quantum)

Quantum & Signal Processing

Azure Quantum, Quantum Development Kit (QDK), EEG-fMRI Signal Processing, Preprocessing & Denoising (RETROICOR, ICA), Feature Extraction

Web, API & Frameworks

Next.js 15, React 19, FastAPI, Flask, Node.js, RESTful API, SOAP (XML Parsing), Tailwind CSS, HTML/CSS

Embedded Systems & Satellite Comm.

RF Telemetry & Telecommand (TM/TC), Link Budget Analysis, Serial Protocols (UART, SPI, I2C)

AI, NLP & Data Science

RAG (Retrieval-Augmented Generation), Prompt Engineering, Scikit-learn, Pandas, NumPy, BERT / TurkishBERT, Zemberek, NLTK, NetworkX, Pyvis (Network Analysis)

Cybersecurity & Cryptography

Post-Quantum Cryptography (PQC/ML-KEM, ML-DSA), liboqs (Open Quantum Safe), Zero Trust (NIST SP 800-207), Ethical Hacking, RSA, AES-GCM

Database & ORM

MSSQL Server, PostgreSQL, Prisma ORM, ChromaDB (Vector DB), Elasticsearch, Database Design & Query Optimization

Tools & DevOps

Docker, Docker Compose, Git, GitHub, Joblib, ReportLab, AFNI/SUMA, EEGLAB, MATLAB, Linux/PowerShell

PERSONAL SKILLS

- Taking Responsibility, Process Tracking, Communication, and Documentation
- Problem Solving and Analytical Thinking
- Project Management and Teamwork

WORK EXPERIENCE

Boğaziçi University - MIMLab (Multimodal Imaging & Mind Lab)

07/2026 – Present

Research and Software Development Intern

Project: MimPhy - Investigation of Cognitive Processes and Physiological Dynamics via Simultaneous EEG-fMRI and Multilayer Network Analysis

- Simultaneous EEG-fMRI & Signal Processing: Preprocessed multimodal neurophysiological data, applied denoising, and extracted features.
- Machine Learning: Developed ML models for cognitive load classification.
- Data Pipelines: Engineered Python/MATLAB-based automated software tools for neuroimaging data analysis.
- Multilayer Network Analysis: Examined the interactions between cognitive processes and physiological dynamics using time-frequency analytics.
- Technologies: Simultaneous EEG-fMRI, Signal Processing, Machine Learning, Feature Extraction, Preprocessing (RETROICOR, ICA), Pipeline Development, Python, MATLAB.

Microsoft

07/2026 – Present

Software Development Intern (Remote)

Project: Quantum Kickstart Program

- Quantum Programming & Q#: Coded and simulated quantum algorithms based on qubit manipulation, superposition, and entanglement principles using Azure Quantum and Q#.
- Hybrid Architectures & SDKs: Analyzed hybrid classical-quantum computing architectures and conducted hands-on problem-solving using modern quantum SDKs.
- Technologies: Q#, Azure Quantum, Quantum Development Kit (QDK), Python, Jupyter Notebooks, Quantum Algorithms.

Presidency of the Republic of Türkiye Directorate of Communications

04/2026 – 06/2026

AI Unit Intern

- Post-Quantum Cryptography (PQC): Migrated public document systems to NIST-standard PQC algorithms (ML-KEM, ML-DSA), performed benchmark analyses, and authored corporate Zero Trust strategy reports.
- AI-Powered RAG & UYAP: Designed a ChromaDB-based RAG architecture using legislative data; integrated UYAP SOAP services to develop a decision support system with KVKK/GDPR-compliant data masking.
- OSINT & Network Analysis: Collected asynchronous data from Telegram via Telethon, performed trend analysis with BERT, and mapped user interaction networks using NetworkX.
- Cyber-Psycho Analysis (SPAS): Developed text-analysis algorithms using TurkishBERT and Zemberek to detect emotional state and radicalization risk, connecting them to a Prisma/MSSQL architecture.
- Technologies: Python, TypeScript, Next.js 15, React 19, FastAPI, Node.js, MSSQL, ChromaDB, PQC (ML-KEM/DSA), liboqs, BERT, Docker, Git.

Uçak Yazılım

01/2026 – 02/2026

Software Development Intern

- Designed and developed a two-version machine learning-based decision support system operating on Finance and HR datasets.
- Developed a Flask-based web application integrating Excel data loading, automatic feature extraction, model selection, and PDF report generation.
- Applied Linear Regression and Logistic Regression algorithms, achieving a 91% accuracy rate in classification tasks.
- Increased system flexibility by building dynamic column detection and automatic model selection mechanisms capable of adapting to varied data schemas.
- Enabled automated textual interpretation of model outputs and structured report generation based on user instructions via LLM integration (Ollama3).
- Improved system stability by creating a data preprocessing pipeline covering missing data (NaN) management, standardization, data validation, and error handling.
- Utilized Joblib for model persistence and ReportLab for automated report generation.
- Designed a modular backend architecture to enhance system sustainability, reusability, and extensibility.
- Technologies: Python, Flask, Pandas, Scikit-learn, HTML, CSS, JavaScript, ReportLab, Joblib, LLM Integration (Ollama3).

PROJECTS

OTUSAT - CubeSat Project

05/2026 – Present

Volunteer Researcher / Software Developer

- Satellite Communications & Telemetry: Currently designing RF communication protocols, telemetry (TM), and telecommand (TC) packet structures between the CubeSat and the ground station.
- Signal Processing & Data Transmission: Developing serial communication (UART, SPI, I2C) architectures and reliable data transmission algorithms among satellite subsystems.

Teknofest Mining Technologies Competition

01/2026 – Present

Team Member

- Team Role: AI Development, Data Analytics & Documentation
- Working on sensor-based data analysis and AI-driven early warning systems within the scope of mining technologies.
- Actively involved in developing machine learning models that detect risky situations in advance by analyzing the rate of change in gas concentration.
- Focusing on enhancing system accuracy and stability by managing data preprocessing, feature extraction, and model optimization processes.
- Managing technical documentation processes to ensure the project maintains a sustainable and clear structure.

Teknofest AI in Aviation Competition

01/2026 – 07/2026

Team Captain

- Actively managed team coordination, task distribution, and project planning processes as Team Captain.
- Led the project architecture design and model development processes for AI-based system engineering.

VOLUNTEER EXPERIENCE

Huawei Student Developers Ankara

02/2026 – Present

Social Media Ambassador

- Management of all social media accounts, content strategy, and boosting community engagement.

Huawei Student Developers Ankara

09/2025 – 02/2026

Secretary

Documentation and communication processes.

Huawei Student Developers Ankara

03/2025 – 11/2025

Organization & Prime Team Lead

Event planning and team management.

OTHER INFORMATION

Languages & Test Scores

- **English:** B2
- **German:** A2
- **ALES** (06.07.2025): 92.8
- **YDS** (05.04.2026): 82.50
- **YÖKDİL** (08.03.2026): 90

REFERENCES

Buğra AYAN, *Head of IT Department*, Presidency of the Republic of Türkiye Directorate of Communications
bugra.ayan@iletisim.gov.tr

Assist. Prof. Dr. Feyza TOKTAŞ, *Project Advisor*, Ankara University, Department of Computer Engineering
ftoktas@ankara.edu.tr

Assoc. Prof. Dr. Yılmaz AR, *Vice Dean*, Ankara University, Faculty of Engineering
yilmazar@gmail.com

TRAININGS AND CERTIFICATIONS

German Language Training (A1-A2) – Certificate of Achievement

- IIENSTITU (International Institute)
- Duration: 30 Hours
- Credential ID: 201090560103
- Verification Code: 2504 BC19 1084 F6D8
- Date: 2019-10

Natural Language Processing

- BTK Academy
- Approved by Information and Communication Technologies Authority
- Cert No: 9XrtzL8DMn
- Date: 2025-12

Machine Learning

- BTK Academy
- Approved by Information and Communication Technologies Authority
- Cert No: qKrhe4vqB2
- Date: 2025-12

HCIA-AI V4.0 Course AI Fundamentals and Applications

- Huawei ICT Academy
- Cert No: ICT20260402000525
- Date: 2026-04

HCCDA-Tech Essentials Course Cloud Developer & Core Technology Essentials

- Huawei ICT Academy
- Cert No: ICT20260402000008
- Date: 2026-04

Defense Industry 401

- Presidency of Defense Industries & YÖK
- Grade: AA (Certificate of Achievement)
- Barcode: UN_0414772827083112410
- Date: 2026-06

Network Fundamentals

- BTK Academy
- Approved by Information and Communication Technologies Authority
- Cert No: xr4tNZmBxo
- Date: 2025-12

Introduction to Deep Learning

- BTK Academy
- Approved by Information and Communication Technologies Authority
- Cert No: eK1hoaBleY
- Date: 2025-12

Complete Ethical Hacking Bootcamp

- Udemy
- Course duration: 28.5 Hours
- Cert No: UC-1aa9aba3-fd7e-4cca-89cc-a91ab86ee733
- Date: 2025-12

HCIA-Cloud Computing V5.5 Course Cloud Computing & Virtualization

- Huawei ICT Academy
- Cert No: ICT20260402001018
- Date: 2026-04

Guidance Systems Speed Bootcamp

- ASELSAN
- Certificate of Participation
- Date: 2026-04

GNÇYTNK'26 Introduction to Professional Life

- Turkcell Academy
- Certificate of Achievement
- Date: 2026-05